Yixuan Xie

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Education

University of Illinois at Urbana-Champaign *08/2026 – Present*
M.S. in Computer Science – Advisor: Prof. Jiawei Han *GPA: 4.0/4.0*

University of Illinois Urbana-Champaign *08/2022 – 05/2026*
B.S. in Computer Science & Statistics – Advisor: Prof. Jiawei Han *GPA: 3.9/4.0*

Publications

† indicates equal contribution.

Retrieval is Cheap, Show Me the Code: Executable Multi-Hop Reasoning for RAG
Jiashuo Sun†, Jimeng Shi†, **Yixuan Xie**, Saizhuo Wang, Jash Rajesh Parekh, Pengcheng Jiang, Zhiyi Shi, Jiajun Fan, Qinglong Zheng, Peiran Li, Shaowen Wang, Ge Liu, Jiawei Han.
arXiv, 2026 [arXiv] [ProjectPage] [Code]

TaSR-RAG: Taxonomy-guided Structured Reasoning for Retrieval-Augmented Generation
Jiashuo Sun†, **Yixuan Xie**†, Jimeng Shi, Shaowen Wang, Jiawei Han.
arXiv, 2026 [arXiv] [Code]

Experience

Amazon-Illinois AICE | *Research Assistant* *05/2026 - Present*
Agent Memory Systems *Champaign, IL*

- Working on memory systems for agents, including memory construction, indexing, updating and retrieval.

UIUC Data Mining Group | *Undergrad Researcher* *08/2025 - 05/2026*
Retrieval-Augmented Generation & RL – Advised by Prof. Jiawei Han *Champaign, IL*

- Built **PyRAG**, reformulated multi-hop RAG as executable Python programs with compiler-grounded self-repair.
- Proposed **TaSR-RAG**, a taxonomy-guided structured reasoning framework for multi-hop QA; +14% over baselines.

Matsuo-Iwasawa Lab | *Research Assistant* *07/2025 - 12/2025*
LLM Post-Training – Mentored by Dr. Qian Niu *Remote*

- Fine-tuned Qwen3 series models with SFT and GRPO, enhanced reasoning on medical QA tasks.

IBM-Illinois Discovery Accelerator Institute | *Research Assistant* *05/2025 - 08/2025*
Open-Schema Information Extraction – Advised by Prof. Jiawei Han *Champaign, IL*

- Benchmarked training-free IE methods and LLM in-context learning across DocEE, MAVE, and NERRE.
- Conducted cross-method error analysis to compare robustness across open-schema IE settings.

DataLynn, Inc. | *Machine Learning Engineer Intern* *05/2024 - 08/2024*
Retrieval-Augmented Generation *Long Island City, NY*

- Designed and deployed a RAG pipeline using Milvus and Zilliz Cloud.
- Launched AI agents for interview prep and customer support, serving 500+ sessions and 2,000+ inquiries.

Selected Projects

Token-Perplexity Signals for Stable RL Post-Training | *Team Lead* *09/2025 - 12/2025*
Machine Learning System Course Project – Mentored by Prof. Fan Lai [Tech Report]

- Investigated gradient-based and token-level diagnostic signals for RLHF dynamics, implementing pipeline instrumentation to capture objective-specific gradients and prefix perplexity during GRPO training.
- Systematically analyzed gradient similarity and perplexity distributions across model scales, revealing limited sensitivity of gradient conflict and substantial overlap in perplexity between correct and incorrect rollouts.

Taxonomy-Enhanced Scientific Retrieval | *Group Lead* *01/2025 - 05/2025*
CS497 Team Project – Advised by Prof. Jiawei Han [Tech Report]

- Investigated taxonomy-enhanced retrieval methods for scientific papers using LitSearch dataset (64K docs).
- Developed and evaluated multiple retrieval strategies—including taxonomy alignment, fusion embedding, and a fine-tuned encoder—demonstrating an improvement in Recall@5 from 39.0% to 47.5% (+8.5%).

Technical Skills

Languages: Python | C/C++
Core Frameworks & Tools: PyTorch | vLLM | veRL | LlamaFactory | Slurm
Systems & Training: ZeRO/DeepSpeed | Retrieval-Augmented Generation | RLHF/RLVR
Others: Docker | MySQL | MongoDB | Kubernetes | FastAPI | Milvus | Flask | Neo4j

Activities & Honors

- **Hackathons** – (2024 & 2025)
- **Datathons** – (2024 & 2025)
- **Course Assistant** – (08/2023 - 12/2023)
- **Dean's List** – (Spring 2025, Fall 2024, Spring 2023)

Last updated: Sep 2, 2026